

1. a) Let U be a random variable which can assume the values 1,2,3,4,5 each with probability .2. Find the c.d.f. of U at 4.5
- b) Given the following c.d.f. find $P\{X = 3\}$.

$$F(x) = 0 \text{ for } x < 1$$

$$F(x) = 0 \text{ for } x < 1$$

$$= \frac{1}{10} \text{ for } 1 \leq x < 3$$

$$= \frac{4}{10} \text{ for } 3 \leq x < 7$$

$$= \frac{9}{10} \text{ for } 7 \leq x < 8$$

$$= 1 \text{ for } x \geq 8$$

2. Let Y_1, Y_2, Y_3 be a random sample of size 3 from a distribution with density

$$f_Y(y) = (\theta + 1)y^\theta, \quad 0 < y < 1, \theta > -1.$$

- a) Find a one-dimensional sufficient statistics.
- b) Find the MLE of θ .
3. A certain type of flashlight is sold with the four batteries included. A random sample of 100 flashlights is obtained and the number of defective batteries in each is determined, resulting in the following data:

Number defective	0	1	2	3	4
Frequency	20	20	20	20	20

Let X be the number of defective batteries in a randomly selected flashlight. Let $p_i = P(i \text{ defectives})$, and consider testing

$$H_0 : p_i = \binom{4}{i} \theta^i (1-\theta)^{4-i}, \quad i = 0, 1, 2, 3, 4.$$

Find the estimated frequency in the category 2 defective.

4. a) Five chips labeled 1 2 3 4 5 are in a hat. Two chips are chosen at random without replacement from the hat. Let S be the sum of the two chips. Find the probability that $S=6$.
- b) On a 3 part exam the probability is $\frac{1}{2}$ of correctly answering the first part. The probability of correctly answering the second part given that the first part is correct is $\frac{3}{4}$; probability of correct on the second part given first part is incorrect is $\frac{1}{4}$. Finally, probability of correctly answering the third part

$\frac{2}{5}$ given both previous parts are missed.

$\frac{1}{5}$ given part 2 is wrong, part 1 is correct.

$\frac{4}{5}$ given part 1 is wrong, part 2 is correct.

$\frac{3}{5}$ given part 1 is correct, part 2 is correct.

Find the probability that only part 2 is correct.